

Leo J. Paggen

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Education

M.S. Data Science

Maastricht University, Netherlands

Sep 2024 – Sep 2026

- **Relevant Coursework:** Machine Learning, Statistical Modeling, Time Series Analysis, Optimization, Natural Language Processing, Algorithms for Big Data

B.S. Economics

Maastricht University, Netherlands

Sep 2020 – Sep 2023

- **Relevant Coursework:** Time Series Forecasting, Statistics, AB Testing

Experience

Amazon - Data Scientist Intern

Seattle, WA

Jul 2025 – Oct 2025

- Extended a production demand forecasting system supporting approximately \$48M in annual cost savings for North American fulfillment operations.
- Owned research initiatives in uncertainty estimation, recurring event modeling, and operational disruption modeling, evaluating improvements using Weighted Absolute Percentage Error (WAPE).
- Built a bootstrap uncertainty framework with multiprocessing, achieving up to $4.7\times$ speedup, and developed holiday and yard-closure forecasting overlays reducing WAPE by up to 6.8%.
- Extended Redshift/SQL data pipelines, authored a 10-page internal research paper, and presented findings to scientists, engineers, and non-technical stakeholders.

PwC - Data Consultant Intern

Brussels, Belgium

Nov 2025 – Jan 2026

- Developed an AI-assisted classification pipeline mapping product descriptions to 4-digit Combined Nomenclature codes for EU Deforestation Regulation compliance.
- Evaluated embedding retrieval and frontier LLM approaches, selecting the strongest architecture through comparative experiments.
- Designed a two-stage LLM validation pipeline that improved recall from 80% to 88% on internal evaluation data.

Maastricht University - Teaching Assistant

Maastricht, NL

Jan 2025 – Present

- Supported courses in Natural Language Processing, Principles of Programming Languages, and Linear Programming, mentoring students in ML, optimization, debugging, and software engineering.

Personal Projects

Static Verification Framework for PyTorch Tensor Programs

Python, Rust, Z3, Protobuf

- Developing a cross-language static analyzer for PyTorch programs using symbolic constraint solving and SMT-based reasoning.
- Built a Python frontend lowering Python ASTs into a semantic IR with source-span metadata and Protobuf serialization.
- Developing a Rust analysis engine with cross-module resolution and Z3 constraint propagation to detect symbolic tensor-shape inconsistencies.

LLM-Based Maritime Autonomy Platform

Python, C++, vLLM, Mistral, Qwen

- Designing an LLM-assisted autonomy layer for MOOS-IvP that combines natural-language reasoning with deterministic vessel control.
- Built a C++/Python interface streaming vessel state into a ReAct agent that generates COLREG-compliant actions through constrained tools.
- Generated synthetic navigation data and fine-tuned Mistral-7B and Qwen-8B models with SFT for reliable structured command generation.

Cisco ACIBILIZER

Python, Ansible

- Developed an open-source tool that converts Cisco ACI configurations into reproducible Ansible automation.

Technical Skills

Languages: Python, Rust, Java, SQL, C++

Machine Learning: Time Series Forecasting, Statistical Modeling, LLMs, NLP, Supervised Fine-Tuning, Uncertainty Quantification

Systems & Tools: PyTorch, vLLM, Z3 SMT Solver, Static Program Analysis, Protobuf, Redshift, Git